# Yuk Fung Angus Chan, PhD

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## Summary

**Data Scientist (PhD)** & **DP-100 Certified Azure Associate** with **6+ years of experience** in Predictive Analytics and Experimental Design for **high-stakes law enforcement** and public sector stakeholders. Expert in architecting **end-to-end data pipelines** for diverse challenges, including **predictive modelling of user behaviour** and **large-scale spatial clustering**, leveraging **MLflow** to ensure high-fidelity experiment tracking and **reproducible deployment**. Proven track record of translating complex statistical findings into **data-driven policy** and **production-ready ML models**. Proficient in **Python, R, and SQL** to bridge the gap between academic rigour and business impact.

## Technical Skills

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| **Machine Learning & MLOps:** | **XGBoost, Random Forest**, Neural Networks (MLP), Hyperparameter Tuning, SMOTE, **MLflow, Azure Model Registry**. |
| **Languages & Cloud:** | **Python (Pandas, Scikit-learn**), R (dplyr, sf, ggplot2), **SQL, Azure Machine Learning**, Git. |
| **Advanced Analytics:** | Multivariate Experimental Design, **A/B Testing**, **Spatial Clustering (DBSCAN**), Longitudinal Data Analysis, Factor Analysis. |
| **Governance & Tools:** | **GDPR Compliance, Data De-identification**, Streamlit, **Power BI,** MS Office. |

## Projects

**End-to-End Cloud Fraud Detection System (Azure MLOps)** [View Repository]

- Engineered an **end-to-end fraud detection system** on **Azure Machine Learning**, achieving an **AUPRC of 0.88**, significantly outperforming the Logistic Regression baseline (0.73).
- Resolved extreme class imbalance (0.17% minority) using **SMOTE** within automated scikit-learn pipelines, boosting **Recall from 65.3% to 82.6%** while maintaining operational false-positive thresholds.
- Orchestrated a "**Model Tournament**" using **MLflow** for experiment tracking, identifying Random Forest with SMOTE as the champion model, yielding a 15% lift in AUPRC over baseline Logistic Regression.
- Evaluated model performance using a Precision-Recall trade-off analysis; selected a tuned Random Forest to maintain 88% Precision, intentionally minimizing "False Positives" to prevent unnecessary customer friction and transaction blocks.
- Performed systematic Hyperparameter Tuning using RandomizedSearchCV to optimise XGBoost and Random Forest architectures, focusing on regularisation parameters to reduce overfitting.
- Executed a **production-ready workflow** by versioning champion models in the **Azure ML Model Registry**, ensuring seamless **model lineage and auditability** for deployment.

**Patient Surge Forecasting Dashboard (NHS Hackathon)**

- Engineered a **Multi-layer Perceptron (MLP)** neural network to forecast patient surges with **85% accuracy**, utilising high-fidelity synthetic data provided by the **NHS Data Science team**.
- Deployed an **automated Streamlit dashboard** to predict resource shortages **10 days in advance**, combining a rule-based scoring system to optimise patient discharge pathways.

## Professional Experience

**Korean National Police Agency** — Remote

**Quantitative Research Consultant** — May 2025 – Present

- Spearheaded the development of **automated end-to-end ETL pipelines** in R, transforming raw multi-source survey data into validated predictive KPIs while **reducing manual processing overhead by 80%.**
- Performed **Exploratory Data Analysis (EDA)** to uncover behavioural patterns in victim data; addressed non-normal distributions (ceiling effects) through advanced transformation techniques, ensuring accuracy of predictive KPIs.
- Established **standardised Git version control**, accelerating collaborative analysis cycles by 2 weeks and and **reducing onboarding time by 40%.**
- Developed **data collection protocols** designed to support multi-wave analysis, ensuring participant IDs remained consistent for future **longitudinal tracking** of satisfaction trends.

**University College London** — London, UK

**Doctoral Researcher** — Sep 2021 – Jan 2026

- Spearheaded a 4-year multi-methodological research program, synthesising diverse datasets including a **large-scale meta-analysis (n=200,966)** and controlled experiments to generate **policy-driven insights**.

- Developed **reproducible analytical pipelines** in R and Python, to automate the cleaning, transformation, and integration of heterogeneous data sources, using Git for version to **reduce manual audit overhead** and maintain data integrity.
- Architected a **custom database and systematic coding schema** to ensure interoperability of complex variables **across 123 international studies**, facilitating large-scale comparative analysis.
- Designed and executed **controlled experiments (n=300)** to quantify the impact of procedural variables on public legitimacy, utilising statistical modelling to identify **predictive behavioural drivers**.
- Translated complex statistical results into **actionable recommendations** for international conferences and executive briefings, resulting in publications in **high-impact journals** including the Journal of Experimental Criminology.
- Established rigorous **GDPR-compliant protocols** and **de-identification workflows** for sensitive qualitative and experimental data, ensuring 100% adherence to ethical mandates.

**Postgraduate Teaching Assistant (Data Science & Statistics)** Sep 2022 – Sep 2025

- Facilitated weekly technical workshops for **100+ students**, covering **geospatial data analysis** and visualisation in R (dplyr, sf, ggplot2).
- Instructed **100+ students** in quantitative research methodologies, providing technical guidance on statistical inference, **hypothesis testing**, and **statistical modelling** on STATA.
- Standardised evaluation process for data projects by developing **automated grading scripts**, ensuring marking consistency and **reducing turnaround time by 20%.**
- Led weekly diagnostic sessions and comprehensive **code reviews** for **100+ end-of-module projects**; audited end-to-end pipelines including data cleaning, feature engineering, and model evaluation.
- Collaborated with Senior Faculty to design course modules, integrating **real-world datasets** to enhance student engagement with **predictive modelling frameworks**.

**Quantitative Research Consultant** Jan 2023 – Jan 2024

- Directed the **full data science lifecycle** in Python, from initial data ingestion and automated cleaning to model validation and **secure data archival**, ensuring 100% compliance with **GDPR and FAIR data principles**.
- Served as **the primary technical liaison** between UCL research team and external industry partners, translating complex statistical methodologies into **actionable policy recommendations** via executive presentations.
- Applied **advanced regression analysis** and hypothesis testing (t-tests) to identify key drivers of user satisfaction, leading to service optimisations that **increased satisfaction by 30%.**
- Designed a complex **multivariate (2x2x2x2 factorial) experiment** and a **scalable data schema** to support multi-factor analysis across 16 experimental cells.
- Engineered **thematic mining workflows** for textual analysis of **unstructured user feedback**, identifying recurring pain points to drive **UX optimisation**.

**City University of Hong Kong** Hong Kong

**Research Assistant, (Full-time)** Sep 2019 – Aug 2021

- Engineered a robust **Data Linkage system** to integrate disparate longitudinal datasets across multiple waves; created a Unique Identifier (UID) framework to maintain 100% record integrity and **eliminated data attrition** for a cohort of 3,000+ participants.
- Developed a **systematic data cleaning pipeline** to identify and resolve anomalies in longitudinal survey responses; applied **data imputation techniques** to handle missing values, ensuring statistical power of final analysis.
- Engineered **composite features** from raw behavioural data by synthesising multiple survey variables into validated **psychometric scales**, improvingthe interpretability of regression models.
- Partnered with cross-functional stakeholders (social workers) to engineer a **standardised screening tool**; applied Factor Analysis and **Reliability Testing** in SPSS to validate instrument's longitudinal consistency and predictive power.
- Co-authored a research grant proposal (HK Beat Drug Fund), creating the **Data Management Plan (DMP)** and methodological framework to assure scalable and reproducible data collection.
- Executed rigorous **data de-identification and anonymisation** of sensitive datasets for peer-reviewed publication, ensuring **100% compliance with privacy mandates** while preserving data utility for complex analysis.

## Certifications

**DP-100: Azure Data Scientist Associate – Microsoft**

IBM Data Science Professional Certificate - Coursera

## Education

**University College London (UCL) | PhD in Security and Crime Science | Jan 2026**
University of Hong Kong | Master of Social Sciences (Criminology) | 2019
University of Hong Kong | Bachelor of Social Sciences (Psychology) | 2018